

CHIRAL JACOBI–CAYLEY DYNAMICS: CYCLOTOMIC S -UNIT OBSTRUCTIONS, EXACT NONPERIODICITY, AND QR RELAXATION

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ABSTRACT. Let J be a real symmetric zero-diagonal Jacobi matrix and let B be the real skew-symmetric tridiagonal matrix obtained by orienting the same edge weights. A diagonal unitary gauge gives the exact identity $B = iS^*JS$; hence the positive eigenvalues of J are precisely the positive rotation frequencies of B . For the scalar Cayley family

$$\mathcal{C}_h(B) = \left(I - \frac{h}{2}B\right) \left(I + \frac{h}{2}B\right)^{-1},$$

the j th rotation block has angle $-2\arctan(h\lambda_j/2)$. Finite order therefore forces every frequency ratio to be a quotient of tangents of rational multiples of π .

We prove an arithmetic obstruction for such quotients. The odd non-Archimedean support of a single rational-angle tangent contains at most one rational prime; consequently the support of a quotient of two such tangents contains at most two. For the canonical four-dimensional chain with

$$\beta_k^2 = \frac{k(k+2)}{4k^2-1}, \quad k = 1, 2, 3,$$

the positive-frequency ratio is

$$r = \frac{\sqrt{21}(103 + 2\sqrt{1471})}{315}, \quad 4725r^4 - 32986r^2 + 4725 = 0.$$

The displayed quartic is primitive and irreducible over $\mathbb{Q}$. Its odd support is exactly $\{3, 5, 7\}$. Thus r is not a quotient of two rational-angle tangents, and no nonzero scalar Cayley step has finite matrix order or finite projective order. An explicit centralizer argument then shows that the conjugation orbit of the canonical Jacobi carrier is never periodic. The field $\mathbb{Q}(r) = \mathbb{Q}(\sqrt{21}, \sqrt{1471})$ has conductor 123564, yielding an independent but weaker cyclotomic denominator obstruction.

We also separate the fixed-generator Cayley system from the state-dependent unshifted QR/Toda map. For the same four-dimensional chain, QR iteration approaches the block-normal set at the exact asymptotic rate λ_-/λ_+ and cannot be periodic. Finally, the real orthogonal isospectral orbit is shown to be Lagrangian inside the ambient unitary coadjoint orbit; the restricted Kirillov–Kostant–Souriau form therefore vanishes identically. Exact symbolic checks and falsification controls accompany the paper.

1. INTRODUCTION

A real skew-symmetric matrix generates rotations, while a real symmetric Jacobi matrix encodes a one-dimensional weighted chain. When the two matrices are built from the same edge weights, the relation between them is stronger than an analogy: a diagonal phase gauge converts one into the other exactly. This observation turns the eigenvalues of a chiral Jacobi chain into rotation frequencies and makes the Cayley transform a natural discrete-time propagator.

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The scalar Cayley transform is attractive because it preserves orthogonality exactly and uses only one real step parameter. That same scalar restriction creates an arithmetic problem. If a skew generator has positive frequencies $\lambda_1, \dots, \lambda_k$, then the Cayley angles are

$$\phi_j(h) = 2 \arctan\left(\frac{h\lambda_j}{2}\right).$$

Exact finite order requires all $\phi_j(h)$ to be rational multiples of 2π . Eliminating h forces every ratio λ_j/λ_1 to be a quotient of tangents of rational angles. For $k \geq 2$, exact scalar periodicity is therefore exceptional even before any specific arithmetic structure is imposed.

The paper develops two layers of obstruction. The first is field-theoretic: a rational-angle tangent quotient lies in a cyclotomic field, so the conductor of the field generated by a frequency ratio must divide a common angle denominator. The second is local and stronger: a single rational-angle tangent can have nonzero valuation at no more than one odd rational prime. A quotient of two tangents can therefore involve no more than two odd primes in its non-Archimedean support.

For the canonical four-dimensional Jacobi chain considered here, the frequency ratio generates the biquadratic field

$$\mathbb{Q}(\sqrt{21}, \sqrt{1471})$$

and has nonzero valuations above each of 3, 5, and 7. The local obstruction therefore excludes every possible rational-angle tangent representation, at every denominator. This closes the scalar commensurability question completely rather than merely excluding a finite range of denominators.

Jacobi matrices and their spectral theory are classical; see Teschl [6]. The Cayley transform originates with Cayley [1]. Spectral and inverse problems for skew-symmetric tridiagonal matrices have a substantial literature, including Fernando [2] and the recent work of Zeng, Xu, and Bebiano [9]. The cyclotomic-unit facts used below are standard and may be found in Washington [7]. The contribution here is the exact assembly of these ingredients into a scalar-Cayley nonperiodicity theorem for a specific chiral Jacobi chain, together with a general prime-support criterion.

1.1. Main results. The principal statements are as follows.

- R1.** The symmetric Jacobi matrix J and its oriented skew companion B satisfy $B = iS^*JS$. Hence $\text{spec}(B) = i\text{spec}(J)$ exactly.
- R2.** A positive frequency vector admits a nonzero scalar Cayley step of finite order precisely when it lies on a rational-angle tangent ray. For two or more frequencies, the set of such vectors is a countable union of rays and is meagre and Lebesgue-null.
- R3.** The odd prime support of $\tan(\pi a/N)$ has cardinality at most one; a quotient of two such tangents has support cardinality at most two.
- R4.** The canonical four-dimensional frequency ratio has monic minimal polynomial

$$x^4 - \frac{32986}{4725}x^2 + 1,$$

equivalently the primitive integral polynomial $4725x^4 - 32986x^2 + 4725$, and odd prime support exactly $\{3, 5, 7\}$.

- R5.** A four-site chain whose frequency ratio has odd support of cardinality at least three has no finite scalar Cayley order; its conjugation orbit is injective. The canonical chain satisfies this criterion for every nonzero step.
- R6.** The generated number field is biquadratic of conductor 123564. This gives an independent necessary denominator divisibility condition, although the local support theorem already proves universal nonexistence.

- R7.** The state-dependent unshifted QR/Toda iteration is a different dynamical system: its cross-block coupling decays at rate λ_-/λ_+ , so it relaxes toward a two-block normal form rather than producing a periodic orbit.
- R8.** The real orthogonal isospectral orbit is a Lagrangian real form of the unitary coadjoint orbit; the restricted KKS two-form vanishes identically.

2. CHIRAL JACOBI PAIRS

2.1. The symmetric and skew chains. Let $m \geq 2$ and let $b_1, \dots, b_{m-1}$ be nonzero real numbers. Define

$$(1) \quad J = \begin{pmatrix} 0 & b_1 & & & \\ b_1 & 0 & b_2 & & \\ & b_2 & 0 & \ddots & \\ & & \ddots & \ddots & b_{m-1} \\ & & & b_{m-1} & 0 \end{pmatrix},$$

and orient the same edges to obtain

$$(2) \quad B = \begin{pmatrix} 0 & -b_1 & & & \\ b_1 & 0 & -b_2 & & \\ & b_2 & 0 & \ddots & \\ & & \ddots & \ddots & -b_{m-1} \\ & & & b_{m-1} & 0 \end{pmatrix}.$$

Thus $J^T = J$ and $B^T = -B$.

Theorem 2.1 (Exact phase-gauge equivalence). *Let*

$$S = \text{diag}(1, i, i^2, \dots, i^{m-1}).$$

Then

$$(3) \quad B = iS^*JS.$$

Consequently

$$(4) \quad \text{spec}(B) = i \text{ spec}(J)$$

with multiplicity.

Proof. For an upper off-diagonal entry,

$$(S^*JS)_{j,j+1} = i^{-(j-1)}b_ji^j = ib_j,$$

whereas the corresponding lower entry is $-ib_j$. Multiplication by i gives $-b_j$ above the diagonal and b_j below it, which is exactly B . $\square$

Let

$$\Sigma = \text{diag}(1, -1, 1, -1, \dots).$$

Then $\Sigma J \Sigma = -J$ and $\Sigma B \Sigma = -B$. The matrices are therefore chiral, and their nonzero spectra occur in opposite pairs.

Corollary 2.2 (Rotation-plane decomposition). *Assume $b_j \neq 0$ for every j . The eigenvalues of J are simple. There are $k = \lfloor m/2 \rfloor$ distinct positive eigenvalues*

$$\lambda_1 > \lambda_2 > \dots > \lambda_k > 0,$$

and an additional zero eigenvalue exactly when m is odd. In a real orthogonal basis,

$$(5) \quad B \sim \bigoplus_{j=1}^k \lambda_j K \quad \text{with} \quad K = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

plus one zero block when m is odd.

Proof. An irreducible real symmetric tridiagonal matrix has simple spectrum. Chirality pairs each nonzero eigenvalue with its negative, and odd dimension forces one unpaired zero. The real canonical form of a skew-symmetric matrix [4] then gives (5), with frequencies identified by theorem 2.1. $\square$

3. SCALAR CAYLEY DYNAMICS AND EXACT FINITE ORDER

For $h \in \mathbb{R}$, define the scalar Cayley transform

$$(6) \quad \mathcal{C}_h(B) = \left(I - \frac{h}{2}B\right) \left(I + \frac{h}{2}B\right)^{-1}.$$

Because B is real skew-symmetric, $I + hB/2$ is invertible for every real h , and $\mathcal{C}_h(B) \in \text{SO}(m)$.

Proposition 3.1 (Cayley angle law). *On the j th rotation plane in (5), $\mathcal{C}_h(B)$ is a rotation through angle $-\phi_j(h)$, where*

$$(7) \quad \phi_j(h) = 2 \arctan\left(\frac{h\lambda_j}{2}\right).$$

For $h > 0$, one has $0 < \phi_j(h) < \pi$.

Proof. On the complex eigenline of B with eigenvalue $i\lambda_j$, the Cayley eigenvalue is

$$\frac{1 - ih\lambda_j/2}{1 + ih\lambda_j/2} = \exp\left(-2i \arctan \frac{h\lambda_j}{2}\right).$$

The conjugate eigenline gives the opposite angle. $\square$

Since $\mathcal{C}_{-h}(B) = \mathcal{C}_h(B)^{-1}$, finite-order questions may be reduced to $h > 0$.

Theorem 3.2 (Ordinary finite-order criterion). *Let $h > 0$. The matrix $\mathcal{C}_h(B)$ satisfies $\mathcal{C}_h(B)^N = I$ if and only if there exist integers*

$$1 \leq a_j < \frac{N}{2}, \quad j = 1, \dots, k,$$

such that

$$(8) \quad \frac{h\lambda_j}{2} = \tan \frac{\pi a_j}{N}.$$

Equivalently, the same integers satisfy, for every j ,

$$(9) \quad \frac{\lambda_j}{\lambda_1} = \frac{\tan(\pi a_j/N)}{\tan(\pi a_1/N)}.$$

and the scale condition $h = 2 \tan(\pi a_1/N)/\lambda_1$. The ratio equations alone characterize the existence of a suitable positive step, not finite order at an arbitrary prescribed step. The exact matrix order is

$$\text{lcm}_{1 \leq j \leq k} \frac{N}{\gcd(N, a_j)}.$$

Proof. By proposition 3.1, the j th plane closes after N steps exactly when $N\phi_j(h) \in 2\pi\mathbb{Z}$. Since $0 < \phi_j(h) < \pi$, write $\phi_j(h) = 2\pi a_j/N$ with $1 \leq a_j < N/2$. Taking the tangent of the half-angle gives (8). The order formula is the least common multiple of the individual rotation orders. $\square$

For conjugation dynamics, the center $\{\pm I\}$ is invisible in even dimension. This motivates the following.

Definition 3.3. For an orthogonal matrix U in even dimension, its projective order is

$$\text{pord}(U) = \min\{q \geq 1 : U^q \in \{I, -I\}\},$$

when such a q exists.

Proposition 3.4 (Projective finite-order criterion). *Assume m is even. For $h > 0$, $\mathcal{C}_h(B)^q = \varepsilon I$, $\varepsilon \in \{\pm 1\}$, if and only if there exist integers m_j with $1 \leq m_j < q$, all of one common parity, such that*

$$(10) \quad \frac{h\lambda_j}{2} = \tan \frac{\pi m_j}{2q}, \quad m_j \equiv \begin{cases} 0 & (\text{mod } 2), \quad \varepsilon = 1, \\ 1 & (\text{mod } 2), \quad \varepsilon = -1. \end{cases}$$

In particular, every frequency ratio is again a quotient of tangents of rational multiples of π .

Proof. The identity $R(q\phi_j) = \varepsilon I_2$ is equivalent to $q\phi_j \equiv 0 \pmod{2\pi}$ when $\varepsilon = 1$ and $q\phi_j \equiv \pi \pmod{2\pi}$ when $\varepsilon = -1$. Since $0 < \phi_j < \pi$, one may write $\phi_j = \pi m_j/q$ with $1 \leq m_j < q$ and the stated parity, and then take half-angle tangents. Conversely, the range $1 \leq m_j < q$ places $\pi m_j/(2q)$ in $(0, \pi/2)$, so the principal arctangent recovers $\phi_j = \pi m_j/q$ and hence $R(q\phi_j) = \varepsilon I_2$. $\square$

3.1. Generic nonperiodicity.

Theorem 3.5 (The scalar finite-order locus is thin). *Fix $k \geq 2$. In the positive frequency cone $\mathbb{R}_{>0}^k$, the set of frequency vectors admitting a nonzero scalar Cayley step of finite ordinary order is a countable union of one-dimensional rays. The same holds for finite projective order. Hence both loci are meagre and have Lebesgue measure zero.*

Proof. For fixed N and integer vector $a = (a_1, \dots, a_k)$, (8) requires

$$(\lambda_1, \dots, \lambda_k) = c \left(\tan \frac{\pi a_1}{N}, \dots, \tan \frac{\pi a_k}{N} \right), \quad c > 0.$$

There are countably many choices of N and a . The projective case is identical with denominator $2q$ and a parity restriction. A countable union of one-dimensional subsets is null and meagre in dimension at least two. $\square$

Example 3.6 (Dimension three is exactly controllable). When $m = 3$, there is only one positive frequency λ . For any $N \geq 3$ and any a coprime to N with $1 \leq a < N/2$,

$$h = \frac{2}{\lambda} \tan \frac{\pi a}{N}$$

produces exact order N . The obstruction begins only when more than one independent rotation plane must be controlled by the same scalar.

4. CYCLOTOMIC UNITS AND ODD PRIME SUPPORT

The finite-order criterion converts dynamics into arithmetic. We now isolate a local obstruction that applies to any quotient of rational-angle tangents.

Definition 4.1 (Odd prime support). Let $\alpha \in \overline{\mathbb{Q}}^\times$. Choose a number field L containing α . Define

$$\text{Supp}_{\text{odd}}(\alpha) = \{p \text{ odd prime} : v_{\mathfrak{P}}(\alpha) \neq 0 \text{ for some } \mathfrak{P} \mid p \text{ in } L\}.$$

This set is independent of the choice of L , since valuations multiply by positive ramification indices under field extension.

We use the standard values of cyclotomic polynomials at ± 1 ; see, for example, Washington [7].

Lemma 4.2 (Cyclotomic unit test). *Let z be a root of unity of exact order $m > 2$.*

- (i) *The algebraic integer $1 - z$ has nonempty odd prime support exactly when $m = p^e$ for an odd prime p ; in that case its odd support is $\{p\}$.*
- (ii) *The algebraic integer $1 + z$ has nonempty odd prime support exactly when $m = 2p^e$ for an odd prime p ; in that case its odd support is $\{p\}$.*

Proof. Since z is primitive of order m ,

$$\left| N_{\mathbb{Q}(z)/\mathbb{Q}}(1 - z) \right| = |\Phi_m(1)|, \quad \left| N_{\mathbb{Q}(z)/\mathbb{Q}}(1 + z) \right| = |\Phi_m(-1)|.$$

The classical evaluations are

$$|\Phi_m(1)| = \begin{cases} p, & m = p^e \text{ for a prime } p, \\ 1, & m \text{ is not a prime power,} \end{cases}$$

and, for $m > 2$,

$$|\Phi_m(-1)| = \begin{cases} p, & m = 2p^e \text{ for an odd prime } p, \\ 2, & m = 2^e, \\ 1, & \text{otherwise.} \end{cases}$$

An algebraic integer of norm ± 1 is a unit. In each nonunit odd-prime case the absolute norm is a power of the single rational prime displayed, so no other odd prime can occur in the support. $\square$

Theorem 4.3 (One-prime support of a rational-angle tangent). *Let $a \in \mathbb{Z}$ and $N \in \mathbb{N}$. Assume $\tan(\pi a/N)$ is finite and nonzero. Then*

$$(11) \quad \left| \text{Supp}_{\text{odd}} \left(\tan \frac{\pi a}{N} \right) \right| \leq 1.$$

More precisely, let $m = N / \gcd(N, a)$ be the exact order of ζ_N^a .

- *If $m = p^e$ for an odd prime p , the odd support is $\{p\}$.*
- *If $m = 2p^e$ for an odd prime p , the odd support is $\{p\}$.*
- *In every other case, the odd support is empty.*

Proof. Put $z = \zeta_N^a$. The tangent identity

$$(12) \quad \tan \frac{\pi a}{N} = -i \frac{z - 1}{z + 1}$$

expresses the tangent as a root of unity times a quotient of the two factors in lemma 4.2. For odd primes, at most one of the numerator and denominator can be a nonunit, and the listed cases follow. $\square$

Corollary 4.4 (Two-prime quotient obstruction). *For any two finite nonzero rational-angle tangents t_1, t_2 ,*

$$(13) \quad |\text{Supp}_{\text{odd}}(t_1/t_2)| \leq 2.$$

Consequently, an algebraic number with nonzero valuations above three distinct odd rational primes cannot be a quotient of two tangents of rational multiples of π .

Proof. The support of a quotient is contained in the union of the supports of its numerator and denominator. Apply theorem 4.3. $\square$

Corollary 4.5 (General scalar-Cayley obstruction). *If a real skew-symmetric generator has two positive frequencies λ_i, λ_j with algebraic ratio λ_i/λ_j such that*

$$|\text{Supp}_{\text{odd}}(\lambda_i/\lambda_j)| \geq 3,$$

then no nonzero scalar Cayley step has finite ordinary order. In even dimension it also has no finite projective order.

Proof. Combine theorem 3.2, proposition 3.4, and corollary 4.4. $\square$

5. THE CANONICAL FOUR-DIMENSIONAL CHAIN

We now specialize to the coefficient sequence

$$(14) \quad \beta_k = \sqrt{\frac{k(k+2)}{4k^2-1}}, \quad k = 1, 2, 3.$$

Thus

$$(15) \quad \beta_1^2 = 1, \quad \beta_2^2 = \frac{8}{15}, \quad \beta_3^2 = \frac{3}{7}.$$

Let J_4 and B_4 be the matrices (1) and (2) with these coefficients; we refer to this specified pair as the canonical four-site chain.

5.1. Exact frequency algebra.

Proposition 5.1 (Squared frequencies). *The characteristic polynomial of J_4 is*

$$(16) \quad \det(xI - J_4) = x^4 - \frac{206}{105}x^2 + \frac{3}{7}.$$

Its positive eigenvalues $\lambda_+ > \lambda_- > 0$ satisfy

$$(17) \quad \lambda_{\pm}^2 = \frac{103 \pm 2\sqrt{1471}}{105}.$$

Proof. For a zero-diagonal four-site chain with weights b_1, b_2, b_3 , continuant expansion gives

$$\det(xI - J) = x^4 - (b_1^2 + b_2^2 + b_3^2)x^2 + b_1^2b_3^2.$$

Substitution of (15) yields (16). Solving the quadratic equation for x^2 gives (17), because

$$206^2 - 4 \cdot 105 \cdot 45 = 16 \cdot 1471.$$

□

Theorem 5.2 (Frequency ratio, minimal polynomial, and field). *Let*

$$r = \frac{\lambda_+}{\lambda_-}.$$

Then

$$(18) \quad r = \frac{\sqrt{21}(103 + 2\sqrt{1471})}{315} = 2.6143555881447318662\dots,$$

and r has monic minimal polynomial

$$(19) \quad m_r(x) = x^4 - \frac{32986}{4725}x^2 + 1.$$

Equivalently, its primitive integral polynomial is

$$(20) \quad 4725x^4 - 32986x^2 + 4725.$$

Moreover

$$(21) \quad r + r^{-1} = \frac{206\sqrt{21}}{315}, \quad r - r^{-1} = \frac{4\sqrt{30891}}{315},$$

and therefore

$$(22) \quad \mathbb{Q}(r) = \mathbb{Q}(\sqrt{21}, \sqrt{1471}).$$

This is a biquadratic extension of degree four with Galois group $C_2 \times C_2$.

Proof. From (17),

$$r^2 = \frac{103 + 2\sqrt{1471}}{103 - 2\sqrt{1471}} = \frac{(103 + 2\sqrt{1471})^2}{4725},$$

because $103^2 - 4 \cdot 1471 = 4725$. Since $315^2 = 21 \cdot 4725$, the positive square root is (18). Direct elimination gives (19), equivalently (20); one may also write

$$r^2 + r^{-2} = \frac{32986}{4725}.$$

The identities (21) follow by direct simplification.

The explicit formula places r in $\mathbb{Q}(\sqrt{21}, \sqrt{1471})$. Conversely, (21) places $\sqrt{21}$ and $\sqrt{30891}$ in $\mathbb{Q}(r)$, and

$$\sqrt{1471} = \frac{\sqrt{21}\sqrt{30891}}{21}.$$

The squarefree radicands 21 and 1471 are independent (indeed, 1471 is prime), so the field has degree four. Hence the monic quartic in (19) is the minimal polynomial over $\mathbb{Q}$, and (20) is its primitive integral multiple. $\square$

5.2. The conductor obstruction.

Proposition 5.3 (Biquadratic conductor). *The three quadratic subfields of $K = \mathbb{Q}(r)$ have discriminants*

$$(23) \quad 21, \quad 5884 = 4 \cdot 1471, \quad 123564 = 4 \cdot 30891.$$

The conductor of K is

$$(24) \quad f_K = \text{lcm}(21, 5884, 123564) = 123564.$$

Proof. The quadratic subfields are $\mathbb{Q}(\sqrt{21})$, $\mathbb{Q}(\sqrt{1471})$, and $\mathbb{Q}(\sqrt{30891})$. Here 21, 1471, and $30891 = 3 \cdot 7 \cdot 1471$ are squarefree. The first radicand is congruent to 1 modulo 4; the latter two are congruent to 3 modulo 4, giving (23). The conductor of a quadratic field equals the absolute value of its fundamental discriminant. For a biquadratic field, the conductor is the least common multiple of the conductors of its three quadratic characters, which gives (24). $\square$

Corollary 5.4 (Hypothetical denominator divisibility). *If, contrary to the main theorem below, there existed integers a, b and $N \geq 1$ with finite nonzero tangents and*

$$(25) \quad r = \frac{\tan(\pi a/N)}{\tan(\pi b/N)},$$

then 123564 would divide N .

Proof. The factors $-i$ cancel in (12), so the quotient in (25) belongs to $\mathbb{Q}(\zeta_N)$. Since r generates K , one would have $K \subseteq \mathbb{Q}(\zeta_N)$. The conductor divisibility theorem for abelian extensions then gives $f_K \mid N$. $\square$

The conductor condition is exact as a field-embedding statement but does not by itself exclude denominators divisible by 123564. The next theorem provides the decisive element-level obstruction.

5.3. Exact three-prime support.

Theorem 5.5 (Odd prime support of the canonical ratio). *The frequency ratio in (18) satisfies*

$$(26) \quad \text{Supp}_{\text{odd}}(r) = \{3, 5, 7\}.$$

Proof. Let $F = \mathbb{Q}(\sqrt{1471})$. Since $1471 \equiv 3 \pmod{4}$, its ring of integers is $\mathcal{O}_F = \mathbb{Z}[\sqrt{1471}]$. Put

$$\alpha = 103 + 2\sqrt{1471}, \quad \bar{\alpha} = 103 - 2\sqrt{1471}.$$

Then

$$(27) \quad N_{F/\mathbb{Q}}(\alpha) = \alpha\bar{\alpha} = 4725 = 3^3 \cdot 5^2 \cdot 7, \quad r^2 = \frac{\alpha}{\bar{\alpha}}.$$

Because α and $\bar{\alpha}$ are algebraic integers, a rational prime outside $\{3, 5, 7\}$ cannot divide either principal ideal. Hence no other odd prime can occur in the support.

For $p \in \{3, 5, 7\}$ one has $1471 \equiv 1 \pmod{p}$, so p splits in F . Write

$$\mathfrak{p}_p = (p, \sqrt{1471} - 1), \quad \bar{\mathfrak{p}}_p = (p, \sqrt{1471} + 1).$$

Both ideals have norm p . Reduction gives

$$\alpha \equiv 105 \equiv 0 \pmod{\mathfrak{p}_p}, \quad \alpha \equiv 101 \not\equiv 0 \pmod{\bar{\mathfrak{p}}_p}.$$

Conjugating exchanges α with $\bar{\alpha}$ and $\mathfrak{p}_p$ with $\bar{\mathfrak{p}}_p$. Comparing ideal norms with (27) therefore yields the exact factorizations

$$(28) \quad (\alpha) = \mathfrak{p}_3^3 \mathfrak{p}_5^2 \mathfrak{p}_7, \quad (\bar{\alpha}) = \bar{\mathfrak{p}}_3^3 \bar{\mathfrak{p}}_5^2 \bar{\mathfrak{p}}_7.$$

Consequently, as a fractional ideal of F ,

$$(r^2) = \left(\frac{\mathfrak{p}_3}{\bar{\mathfrak{p}}_3}\right)^3 \left(\frac{\mathfrak{p}_5}{\bar{\mathfrak{p}}_5}\right)^2 \left(\frac{\mathfrak{p}_7}{\bar{\mathfrak{p}}_7}\right).$$

After extension to any number field containing r , these valuations remain nonzero. Since $v(r^2) = 2v(r)$, the valuations of r above each of 3, 5, and 7 are nonzero, proving (26). $\square$

Corollary 5.6 (No rational-angle tangent quotient). *There do not exist rational numbers u, v for which both tangents are finite and nonzero and*

$$r = \frac{\tan(\pi u)}{\tan(\pi v)}.$$

Proof. Take a common denominator for u and v . Then apply [corollary 4.4](#) and [theorem 5.5](#). $\square$

6. EXACT SCALAR-CAYLEY NONPERIODICITY

The absence of finite matrix order follows immediately from [corollary 4.5](#). To pass from matrix order to periodicity of the conjugation orbit, we establish that the only powers of the Cayley matrix that can commute with the four-site Jacobi carrier are central.

6.1. A centralizer lemma. Let J and B be the general four-dimensional matrices with nonzero weights b_1, b_2, b_3 . A direct calculation gives

$$(29) \quad [J, B^2] = \begin{pmatrix} 0 & -2b_1b_2^2 & 0 & 0 \\ 2b_1b_2^2 & 0 & 2b_2(b_1^2 - b_3^2) & 0 \\ 0 & 2b_2(-b_1^2 + b_3^2) & 0 & 2b_2^2b_3 \\ 0 & 0 & -2b_2^2b_3 & 0 \end{pmatrix}.$$

In particular, $[J, B^2] \neq 0$ for every irreducible chain.

Lemma 6.1 (Central powers are scalar). *Let J be a four-dimensional irreducible zero-diagonal Jacobi matrix, let B be its oriented skew companion, and let $U = \mathcal{C}_h(B)$ with $h \neq 0$. If U^ℓ commutes with J for some integer $\ell \geq 1$, then*

$$U^\ell \in \{I, -I\}.$$

Proof. The spectrum of J is simple. A real orthogonal matrix commuting with J therefore acts by a scalar ± 1 on each real one-dimensional eigenspace of J . Hence U^ℓ is a symmetric involution.

On the other hand, U^ℓ is a rational function of B . Let P_+ and P_- be the two real rank-two spectral projectors of $-B^2$, associated with the distinct positive frequencies λ_+^2 and λ_-^2 . On each plane U^ℓ is a rotation. Since it is symmetric, that rotation must be I_2 or $-I_2$. Thus

$$U^\ell = \varepsilon_+ P_+ + \varepsilon_- P_-, \quad \varepsilon_\pm \in \{\pm 1\}.$$

If $\varepsilon_+ \neq \varepsilon_-$, then $U^\ell = cI + dB^2$ with $d \neq 0$, because the two projectors are affine polynomials in B^2 . Commutation with J would imply $[J, B^2] = 0$, contradicting (29). Therefore $\varepsilon_+ = \varepsilon_-$ and $U^\ell = \pm I$. $\square$

Theorem 6.2 (Four-site prime-support criterion). *Let J be any irreducible four-dimensional zero-diagonal Jacobi matrix, let B be its oriented skew companion, and let $\lambda_+ > \lambda_- > 0$ be its positive frequencies. Assume the ratio is algebraic and*

$$|\text{Supp}_{\text{odd}}(\lambda_+/\lambda_-)| \geq 3.$$

For every real $h \neq 0$, the matrix $U = C_h(B)$ has infinite ordinary and projective order, and the map

$$\mathbb{Z} \longrightarrow \text{Sym}_4(\mathbb{R}), \quad t \longmapsto U^t J U^{-t},$$

is injective.

Proof. Finite ordinary or projective order would express λ_+/λ_- as a quotient of rational-angle tangents by theorem 3.2 and proposition 3.4, contradicting corollary 4.4. If two orbit states with indices $s < t$ were equal, then U^{t-s} would commute with J . By lemma 6.1, $U^{t-s} = \pm I$, contradicting infinite projective order. $\square$

Corollary 6.3 (Universal scalar nonperiodicity for the canonical chain). *Let J_4 and B_4 be the canonical matrices defined by (14). For every real $h \neq 0$:*

- (i) $C_h(B_4)$ has infinite matrix order;
- (ii) $C_h(B_4)$ has infinite projective order;
- (iii) the conjugation orbit

$$(30) \quad J_t = C_h(B_4)^t J_4 C_h(B_4)^{-t}$$

consists of pairwise distinct matrices and hence has no positive period.

Proof. Apply theorems 5.5 and 6.2. $\square$

Remark 6.4 (What the theorem does and does not say). The orbit (30) is bounded, isospectral, and Bohr almost periodic because it is a finite sum of torus characters. The theorem excludes exact recurrence after a finite integer number of steps. It does not assert rational independence of the two Cayley angles, nor does it identify the dimension of the orbit closure inside the two-torus.

6.2. A periodic control example. The obstruction is not a defect of the Cayley transform itself. It is an arithmetic property of the canonical frequency ratio.

Example 6.5 (An exactly periodic irreducible four-site chain). Set

$$\lambda_+ = \tan \frac{\pi}{5}, \quad \lambda_- = \tan \frac{\pi}{7},$$

and define

$$b_1 = b_3 = \sqrt{\lambda_+ \lambda_-}, \quad b_2 = \lambda_+ - \lambda_-.$$

The resulting zero-diagonal Jacobi matrix has eigenvalues $\pm \lambda_+$ and $\pm \lambda_-$. With $h = 2$, the Cayley angles are $2\pi/5$ and $2\pi/7$, so the Cayley matrix and its conjugation orbit have exact period 35.

Proof. The characteristic polynomial is

$$x^4 - (b_1^2 + b_2^2 + b_3^2)x^2 + b_1^2 b_3^2 = (x^2 - \lambda_+^2)(x^2 - \lambda_-^2).$$

The angle statement follows from (7). Since 5 and 7 are coprime and odd, the matrix order and projective order are both 35. The chain is irreducible, so lemma 6.1 then gives the same exact period for the conjugation orbit. $\square$

7. INDEPENDENT PLANE CONTROL VERSUS SCALAR CONTROL

The scalar no-go should not be confused with an obstruction to periodic orthogonal dynamics on the same rotation planes.

Proposition 7.1 (Exact multi-plane closure). *Let*

$$B = W \left(\bigoplus_{j=1}^k \lambda_j K \oplus 0_\delta \right) W^T, \quad \delta \in \{0, 1\},$$

be the real rotation-plane decomposition. Fix $N \geq 1$ and integers $a_1, \dots, a_k$. Then

$$U = W \left(\bigoplus_{j=1}^k R(2\pi a_j/N) \oplus I_\delta \right) W^T$$

satisfies $U^N = I$. Thus exact closure is always possible when each plane angle is independently controlled.

Proof. Each block closes after N steps. $\square$

The canonical obstruction is therefore a control-rank statement with arithmetic content: one scalar h is required to satisfy two incompatible tangent equations. It does not prohibit exact periodicity under a higher-rank torus control.

8. THE STATE-DEPENDENT QR/TODA MAP

The fixed-generator Cayley orbit and the unshifted QR iteration are both isospectral, but they are not the same dynamical system. The Cayley family holds B fixed. QR iteration recomputes a factorization from the current state.

Let $A_0 = J_4$. At each step take the unique QR factorization

$$A_t = Q_t R_t$$

with positive diagonal in R_t , and define

$$(31) \quad A_{t+1} = R_t Q_t = Q_t^T A_t Q_t.$$

This is the standard unshifted symmetric QR map. It belongs to the QR/Toda factorization setting; every claim below concerns precisely (31), rather than a separately chosen Toda time discretization; see, for example, Teschl [6] and Watkins [8].

8.1. Block convergence. Let the eigenvalues of J_4 be ordered as

$$\lambda_+, -\lambda_+, \lambda_-, -\lambda_-, \quad \lambda_+ > \lambda_- > 0,$$

and set

$$(32) \quad \rho = \frac{\lambda_-}{\lambda_+} = r^{-1} = 0.382503437762896 \dots$$

Theorem 8.1 (Exact QR relaxation rate). *Let A_t be the unshifted QR orbit (31) of the canonical chain. Partition A_t after the first two coordinates:*

$$A_t = \begin{pmatrix} X_t & C_t \\ C_t^T & Y_t \end{pmatrix}, \quad X_t, Y_t, C_t \in \mathbb{R}^{2 \times 2}.$$

There exists a constant $c > 0$ such that

$$(33) \quad \|C_t\|_F = c\rho^t + O(\rho^{2t}).$$

Consequently

$$(34) \quad \lim_{t \rightarrow \infty} \frac{\|C_{t+1}\|_F}{\|C_t\|_F} = \rho.$$

In particular, the orbit approaches the set of symmetric block-diagonal matrices with block spectra $\{\lambda_+, -\lambda_+\}$ and $\{\lambda_-, -\lambda_-\}$.

Proof. Let $\hat{Q}_t = Q_0 Q_1 \cdots Q_{t-1}$ be the accumulated orthogonal factor. Standard QR identities [5, 8] give

$$A_0^t = \hat{Q}_t \hat{R}_t, \quad A_t = \hat{Q}_t^T A_0 \hat{Q}_t,$$

with $\hat{R}_t$ upper triangular. Hence the first two columns of $\hat{Q}_t$ span $A_0^t E$, where $E = (e_1, e_2)$.

Choose an orthogonal eigenbasis so that

$$A_0 = V \begin{pmatrix} \Lambda_+ & 0 \\ 0 & \Lambda_- \end{pmatrix} V^T, \quad \Lambda_+ = \text{diag}(\lambda_+, -\lambda_+), \quad \Lambda_- = \text{diag}(\lambda_-, -\lambda_-).$$

Write $V^T E = (C_+^T, C_-^T)^T$. Both C_+ and C_- are invertible. Indeed, if v is an eigenvector for either positive eigenvalue λ , chirality gives Σv for $-\lambda$; their first two coordinates form

$$\begin{pmatrix} v_1 & v_1 \\ v_2 & -v_2 \end{pmatrix},$$

whose determinant is nonzero because irreducibility gives $v_1 \neq 0$ and the first row of $J_4 v = \lambda v$ gives $v_2 = \lambda v_1 / \beta_1 \neq 0$.

Therefore

$$\begin{aligned} A_0^t E &= V \begin{pmatrix} \Lambda_+^t C_+ \\ \Lambda_-^t C_- \end{pmatrix} \\ &= V \begin{pmatrix} I \\ F_t \end{pmatrix} \Lambda_+^t C_+, \end{aligned}$$

where

$$F_t = \Lambda_-^t M \Lambda_+^{-t}, \quad M = C_- C_+^{-1}.$$

Set $D = \text{diag}(1, -1)$. Since $\Lambda_\pm = \lambda_\pm D$ and $D^2 = I$,

$$(35) \quad F_t = \rho^t D^t M D^t.$$

In particular, $\|F_t\|_2 = \rho^t \|M\|_2$ exactly.

Let P_t be the orthogonal projector onto the first two columns of $\hat{Q}_t$, transported into the eigenbasis of A_0 , and let $P_+ = \text{diag}(I_2, 0)$. The graph representation above gives

$$P_t = P_+ + \begin{pmatrix} O(\|F_t\|_2^2) & F_t^T + O(\|F_t\|_2^3) \\ F_t + O(\|F_t\|_2^3) & O(\|F_t\|_2^2) \end{pmatrix}.$$

Put $\Lambda = V^T A_0 V = \text{diag}(\Lambda_+, \Lambda_-)$. The Frobenius norm of the off-diagonal block C_t is basis-independent and, in these eigenbasis coordinates, equals

$$\|C_t\|_F = \|(I - P_t) \Lambda P_t\|_F.$$

Expanding to first order at P_+ yields

$$(36) \quad \|C_t\|_F = \|\Lambda_- F_t - F_t \Lambda_+\|_F + O(\|F_t\|_2^2).$$

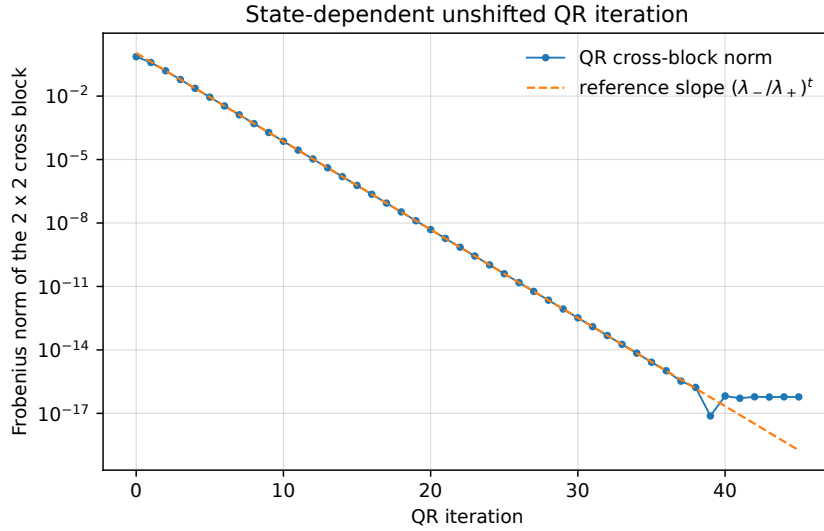


FIGURE 1. The cross-block norm in the state-dependent unshifted QR iteration. The observed asymptotic ratio agrees with $\rho = \lambda_-/\lambda_+$. The plot is illustrative; [theorem 8.1](#) is analytic.

Using [\(35\)](#),

$$\Lambda_- F_t - F_t \Lambda_+ = \rho^t D^t (\lambda_- DM - \lambda_+ MD) D^t.$$

Hence [\(33\)](#) holds with

$$c = \|\lambda_- DM - \lambda_+ MD\|_F.$$

This constant is positive: if it vanished, then $\lambda_- DMD = \lambda_+ M$; taking Frobenius norms and using $M \neq 0$ would force $\lambda_- = \lambda_+$. Finally, [\(34\)](#) follows immediately from [\(33\)](#). $\square$

Corollary 8.2 (The state-dependent QR orbit is not periodic). *The unshifted QR orbit of the canonical chain is not periodic.*

Proof. Initially C_0 contains the nonzero central edge β_2 . By [theorem 8.1](#), $\|C_t\|_F \rightarrow 0$. A periodic nonnegative sequence that converges to zero must be identically zero, a contradiction. $\square$

The QR map therefore relaxes toward a two-block normal form. It does not implement the fixed-generator rotation of [\(30\)](#). The persistent 2×2 blocks are forced by the equal absolute values within each chiral pair $\pm\lambda_\pm$; unshifted QR separates distinct moduli but cannot order the two signs within an equal-modulus pair.

9. THE UNITARY SYMPLECTIC ENVELOPE

This section records the correct geometric placement of the real isospectral orbit. The result is standard in the geometry of coadjoint orbits [\[3\]](#); it is included to fix the scope of any symplectic interpretation.

Let J be a real symmetric matrix with simple spectrum. Its unitary orbit

$$\mathcal{O}_U(J) = \{UJU^* : U \in U(m)\}$$

is a Hermitian coadjoint orbit. Its tangent vectors at J are $[X, J]$ with X anti-Hermitian, and the Kirillov–Kostant–Souriau form is

$$(37) \quad \omega_J([X, J], [Y, J]) = -i \operatorname{tr}(J[X, Y]).$$

The real orthogonal orbit is

$$\mathcal{O}_O(J) = \{QJQ^T : Q \in O(m)\} \subset \mathcal{O}_U(J).$$

Proposition 9.1 (The real orbit is Lagrangian). *For simple real spectrum, $\mathcal{O}_O(J)$ is a Lagrangian submanifold of $\mathcal{O}_U(J)$. In particular, the restriction of (37) to the real orbit is identically zero.*

Proof. For tangent vectors to the real orbit, X, Y are real skew-symmetric. Their commutator $[X, Y]$ is again real skew-symmetric. Since J is real symmetric,

$$\mathrm{tr}(J[X, Y]) = 0,$$

so the KKS form vanishes on $T_J\mathcal{O}_O(J)$. Unitary invariance of the KKS form, restricted to the orthogonal subgroup, gives the same conclusion at every point of $\mathcal{O}_O(J)$.

For simple spectrum,

$$\dim_{\mathbb{R}} \mathcal{O}_U(J) = m^2 - m = m(m - 1),$$

whereas

$$\dim_{\mathbb{R}} \mathcal{O}_O(J) = \frac{m(m - 1)}{2}.$$

Thus the real orbit is isotropic of half the ambient dimension and hence Lagrangian. $\square$

Remark 9.2. A numerical test of $\mathrm{tr}(J[X, Y])$ using real skew-symmetric X and Y will always return zero for algebraic reasons. Such a test cannot certify a nondegenerate symplectic form on the real orbit. The nondegenerate KKS structure lives on the full unitary orbit; the real orthogonal orbit is its Lagrangian real form.

10. COMPUTATIONAL VERIFICATION AND FALSIFICATION CONTROLS

The proofs above are exact and do not depend on numerical experiments. A supplementary Python/SymPy implementation performs the following independent checks.

- (i) It verifies symbolically the gauge identity $B = iS^*JS$ and confirms that a one-edge sign mutation breaks it.
- (ii) It reconstructs (16)–(20), the field identities (21), and the conductor factorization.
- (iii) It checks the residue certificates at $p = 3, 5, 7$ and verifies the values $\Phi_m(\pm 1)$ for every $3 \leq m \leq 500$ against the prime-power classification in lemma 4.2.
- (iv) It performs a monotonicity-reduced numerical scan of tangent quotients through denominator $N = 5000$. The closest observed value differs from r by approximately 4.0×10^{-10} ; this is only a regression test, not evidence for the theorem.
- (v) It constructs the exactly periodic control example in example 6.5 and checks its eigenvalues numerically. The original verifier records the analytically proved order 35 as a constant; the corrected supplement adds high-precision checks of $U^{35} = I$, exclusion of earlier central powers, and closure of the conjugation orbit.
- (vi) It runs the state-dependent QR map and fits an asymptotic decay factor $0.382503318\dots$, agreeing with the exact ratio limit $\rho = 0.382503437\dots$ before roundoff dominates.

The mutation and periodic-control tests check that the implementation detects the altered gauge identity and admits a periodic example when the arithmetic obstruction is absent. Finite-precision controls do not prove exact order or universal nonperiodicity; those conclusions follow from the written proofs. In the 8 October 2026 rerun, the unchanged verifier passed all its regression checks with the pinned package versions under Python 3.12.14. An additional 100-digit implementation checks QR decay beyond the original floating-point range and tests the fixed-step scale condition in theorem 3.2.

11. CONCLUSION

The scalar Cayley transform converts a chiral Jacobi spectrum into a simultaneous rational-angle problem. For the canonical four-dimensional chain, the decisive obstruction is local rather than merely field-theoretic. A quotient of two rational-angle tangents can involve at most two odd rational primes in its non-Archimedean support, while the canonical frequency ratio involves exactly three. This proves universal nonperiodicity for every nonzero scalar step and eliminates the need for an unbounded denominator search.

Three distinctions are structurally important. First, scalar control is much weaker than independent control of the rotation planes; exact periodicity is readily restored on the full torus. Second, fixed-generator Cayley dynamics is not the state-dependent QR/Toda map: the former is nonperiodic almost-periodic rotation, whereas the latter relaxes toward a chiral two-block normal form. Third, the real orthogonal orbit is Lagrangian, not symplectic, inside the unitary coadjoint orbit.

The prime-support criterion applies beyond the canonical chain. Any skew generator with an algebraic frequency ratio supported at three or more odd rational primes is excluded from exact scalar Cayley periodicity. This supplies a practical arithmetic gate for future exact-period designs: before searching over angle denominators, compute the non-Archimedean support of the spectral ratios.

APPENDIX A. EXACT MATRICES AND IDENTITIES

For reference, the canonical matrices are

$$J_4 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & \sqrt{8/15} & 0 \\ 0 & \sqrt{8/15} & 0 & \sqrt{3/7} \\ 0 & 0 & \sqrt{3/7} & 0 \end{pmatrix},$$

$$B_4 = \begin{pmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & -\sqrt{8/15} & 0 \\ 0 & \sqrt{8/15} & 0 & -\sqrt{3/7} \\ 0 & 0 & \sqrt{3/7} & 0 \end{pmatrix}.$$

The squared-frequency polynomial is

$$105y^2 - 206y + 45,$$

whose discriminant is $16 \cdot 1471$. The primitive reciprocal integral polynomial for r follows from

$$r^2 + r^{-2} = \frac{\lambda_+^4 + \lambda_-^4}{\lambda_+^2 \lambda_-^2} = \frac{32986}{4725}.$$

APPENDIX B. REPRODUCIBILITY MANIFEST

The supplementary archive contains an executable symbolic and numerical verifier, a machine-readable JSON summary, numerical tables for the tangent scan and QR relaxation, vector and raster copies of [figure 1](#), and a README recording exact filenames, execution instructions, and dependency versions.

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